Aryan Goel

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MS CS candidate building and deploying ML systems end-to-end — from model training through MLOps, monitoring, and agentic AI serving layers.

EDUCATION

University of Southern California — Los Angeles, CA
Master of Science, Computer Science (Data Science Specialization) — *Aug. 2025 – May 2027 (Expected)*
***Relevant Coursework**: Machine Learning for Data Science, Applied Statistical Inference, Deep Learning, Algorithms, Database Systems (indexing, storage engines, query optimization), Foundations of Data Management (Hadoop, Spark, distributed SQL), NLP*

Manipal University Jaipur — Jaipur, India
Bachelor of Technology, Computer & Communication Engineering — GPA: 9.15/10 — *July 2021 – May 2025*

TECHNICAL SKILLS

Languages: Python, SQL, Core Java
Deep Learning: PyTorch, CNNs, Computer Vision (YOLO, OpenCV, Faster R-CNN — anchor boxes, NMS), Attention Mechanisms, Residual Networks
Classical ML & Search: scikit-learn, XGBoost, SHAP, Isolation Forest/LOF, Uplift Modeling, Minimax/Alpha-Beta, Reinforcement Learning (prototype)
MLOps & Deployment: MLflow, DVC (data/pipeline versioning), Docker, TensorBoard, FastAPI, Postman, CI/CD (GitHub Actions), AWS EC2, Model Versioning
Data & Streaming: PySpark, Kafka, Spark Structured Streaming, ClickHouse, TimescaleDB, Dagster, dbt
GenAI & Agentic Systems: Google ADK, LangGraph, LangMem (persistent memory), Gemini Pro, Multi-Agent Architectures, RAG (retrieval, vector search), LLM API Integration
Dev Tooling & Backend: GitHub API (automated code review), FastAPI backends, Supabase, Databases & Tools: PostgreSQL, SQL, Git, Postman

EXPERIENCE

Indian Institute of Remote Sensing (ISRO) — Dehradun, India
Research Intern — *Feb. 2025 – May 2025*

- Engineered a cascaded CNN architecture (CED-Net) for crop/weed segmentation on RGB drone imagery, outperforming a baseline U-Net through residual blocks and attention mechanisms
- Built the full annotation-to-evaluation pipeline (LabelMe → pixel masks → training/eval on **IoU and Dice score**), benchmarking multiple architecture variants

Indian Institute of Technology (IIT) Roorkee — Roorkee, India
Research Summer Engineer — *May 2024 – July 2024*

- Devised a Physics-Informed Gaussian Process Regression model to forecast flood events, embedding hydrological constraints directly into the model for improved reliability over a standard GPR baseline
- Executed **uncertainty quantification** via Monte Carlo and Multi-Level Monte Carlo simulations for confidence-bounded risk predictions

Indian Institute of Technology (IIT) Roorkee — Roorkee, India
Software Developer — *June 2023 – July 2023*

- Managed and optimized geospatial data layers in **GeoServer** for a production WebGIS application; performed data classification with QGIS
- Redefined the rendering pipeline and resolved performance bugs to reduce load time, while shipping new data-classification features

PROJECTS

ChainSignal: Crypto Market & On-Chain Anomaly Detection | *Python, PySpark, ClickHouse, scikit-learn*

- With no labels, compared a robust z-score, Isolation Forest, and LOF using 887 planted anomalies and a backtest on documented events: Isolation Forest recovered 43% of 12σ combination anomalies vs 11% for z-score at a 1% alert budget; the Kelp DAO hack was missed and reported as a limitation.
- Built the ClickHouse + PySpark pipeline (1,000 coins × 365 days, 1.64M transfers) with Dagster, dbt data-contract tests, and 69 pytest tests.

Little-Go: Adversarial Search Game-Playing AI Agent | *Python, Minimax/Alpha-Beta Pruning, Reinforcement Learning*

- Built a competitive Go-playing agent (5×5 board) using minimax search with alpha-beta pruning and a custom heuristic evaluator, winning 60/60 games in the course's automated grading tournament.
- Validated gains with a 200–400 game tournament vs. the prior version, reporting a 63.5% win rate (95% CI).
- Engineered a 61-test suite via mutation testing (12 planted bugs) to verify real regression coverage.
- Prototyped a standalone RL pipeline (22,000 endgame positions) improving win-margin prediction by up to 5.7 points — a separate experiment not yet folded into the submitted agent.

MarketPulse: Real-Time Trading Anomaly Detector | *Python, Kafka, Spark Structured Streaming, TimescaleDB, Gemini*

- Built a Kafka + Spark Structured Streaming pipeline that scores live stock and crypto trades on log-return, log-volume, and trade-rate z-scores against leakage-free rolling baselines, with a 0–3 severity score and alert cooldown.
- Added a Gemini agent that explains each flagged event; switched from a local Llama 3.2 3B model after it called a 338σ move "normal noise", treating factual correctness of explanations as a requirement.

Telecom Churn & Retention Intelligence | *Python, XGBoost, scikit-learn, SHAP*

- Built a leakage-safe churn pipeline (nested CV, in-pipeline SMOTE, isotonic calibration, 0.18 cost-optimal threshold cutting expected cost by more than half) and re-ran it on real IBM Telco data (ROC-AUC 0.840).
- Estimated heterogeneous treatment effects with S/T/X-learner uplift models evaluated by Qini/AUUC; reported that T-learner ranked better while X-learner estimated magnitude closer to the known synthetic truth.

Object Detection (Faster R-CNN) — Full MLOps Loop | *PyTorch, DVC, TensorBoard, FastAPI, Postman*

- Trained a **Faster R-CNN** object detector with experiment tracking in TensorBoard and **DVC**-managed data/pipeline versioning for full reproducibility across runs
- Deployed the trained model behind a FastAPI serving layer with request validation, tested via Postman collections

ACHIEVEMENTS

- Second Runner-Up, UCLA Glitch × Google DeepMind Hackathon — built BrandSync, then developed it fully after the Google team's positive feedback
- 1st Place, Inter-College Hackathon (Mental Health track) — built a Figma-prototyped app with an integrated chatbot
- Dean's List for Academic Excellence, Manipal University Jaipur (awarded twice)